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Data Cleaning and Pre-Processing Activities

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I. Dataset

In the field of structural design, the quality of a system depends heavily on the information used to guide its decision-making. To build a reliable generative model, we have selected the "Learning to Simulate and Design for Structural Engineering" (LSDSE) dataset as our primary reference. This dataset is a comprehensive collection of 2,800 building designs, each represented as a structural graph that captures detailed elements, connectivity relationships, and performance metrics.

This dataset is an ideal fit for our thesis because it serves as a necessary benchmark for establishing boundaries for our model. A primary challenge in generative design is preventing it from producing building layouts that are physically impossible or impractical. By analyzing the geometric and spatial properties within the LSDSE collection, such as bay widths, floor counts, and story heights, we can identify patterns that represent standard, functional structural configurations.

These findings allow us to determine specific "input ranges" for our generative output. Rather than allowing the system to suggest arbitrary dimensions, we use the data to define the minimum and maximum limits for the design parameters. We use the LSDSE dataset to teach the system the boundaries of acceptable design. This ensures that every new building generated by our model remains within the logical limits, effectively bridging the gap between theoretical generation and practical application.

II. Dataset Preparation

A. Conversion of the Dataset from .json File to Table



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The original dataset consisted of 28 JSON files containing 2,800 graph-based structural building data. Each file stored information about structural elements, connectivity relationships, story assignments, and structural performance metrics in a hierarchical format. While this format is suitable for direct simulation and graph processing, most learning and analytical techniques require data to be organized in a flat tabular structure. Hence, there is a need to extract, transform, clean, and organize the raw JSON data into relational tables suitable for analysis and model development, while preserving the structural relationships and performance information contained in the original data.

1. JSON File Extraction

```
zip_path = "json_files.zip"
extract_folder = "unzipped_jsons"
with zipfile.ZipFile(zip_path, "r") as zip_ref:
    zip_ref.extractall(extract_folder)
```

The dataset was initially provided as a compressed ZIP file containing all 28 JSON files. The archive was then extracted into a working directory to ensure that every JSON file could be systematically accessed and processed. Without this step, each file would need to be manually accessed, leading to inefficiencies and making it difficult to reproduce.

2. File Sorting (Data Organization)

```
json_files.sort(key=lambda f: int(re.findall(r"\d+", os.path.basename(f))[0])
                 if re.findall(r"\d+", os.path.basename(f)) else 0)
```

By default, file systems and Python's sorting mechanism perform lexicographic ordering (based on string value), which would place data10-19.json before data2.json. Such ordering would cause inconsistencies



in the assignment of building identifiers during the later stages because every building in the dataset would be mislabeled. Not to mention that it would make the data difficult to trace back to its source.

Hence, the extracted JSON files were sorted according to the numerical suffixes in their filenames (e.g., data0.json, data1.json, ..., data27.json) using a natural sorting approach. This preserves the intended numerical sequence of the files and guarantees that the structural data are processed in the same order in which they were originally generated.

3. Building ID Generation

```
for file_index, file_path in enumerate(json_files):  
    file_name = os.path.basename(file_path)  
    file_start_building_id = file_index * 100
```

```
for local_b_id, building in enumerate(buildings_list):  
    continuous_building_id = file_start_building_id + local_b_id
```

Every building graph across all 28 files was assigned a globally unique and continuous integer ID (continuous_building_id) using the formula $\text{file_start_building_id} = (\text{file_index} * 100) + \text{local_b_id}$. This formula ensures that every building can be distinguished from one another after the data from all files has been combined: data0.json contains buildings 0-99, data1.json contains buildings 100-199, etc.

This is necessary because the original JSON files reuse building indices; each file contains a building numbered 0, 1, 2, and so forth. Directly merging the files would create identifier collisions because multiple unrelated buildings share the same ID, which in turn causes ambiguities and



makes it impossible to correctly link building-related records across tables. Hence, the generated building ID serves as a primary identifier that connects the structural elements, graph connections, and story-level analytics belonging to the same building.

4. Story-Level Mapping

```
story_mapping = {}
for story_idx, element_ids in enumerate(stories):
    for eid in element_ids:
        story_mapping[eid] = story_idx + 1
```

Originally, the `bar_id_in_stories` attribute stores story membership information as nested lists, wherein each list contains the elements belonging to a particular floor. To simplify analysis, a lookup dictionary was created to directly map and label each structural element to its corresponding story level. This transformation makes building story information immediately accessible instead of repeatedly traversing nested lists.

5. Structural Element Extraction and Data Flattening

```
coords_arr = bar_nodes.get("end_point_locations", [])
if_beam = bar_nodes.get("if_beam", [])
if_roof = bar_nodes.get("if_roof_beam", [])
if_boundary = bar_nodes.get("if_boundary_beam", [])
cross_sec = bar_nodes.get("cross_section", [])
deck_area = bar_nodes.get("metal_deck_area_on_beam", [])
```

```
element_row = {
    "global_element_id": clean_global_id,
    "story_level": story_mapping.get(i, None),
    "start_x": coords[0], "start_y": coords[1], "start_z": coords[2],
    "end_x": coords[3], "end_y": coords[4], "end_z": coords[5],
    ...
}
```



```
}
```

Each structural element stored within the nested JSON structure was flattened or transformed into a single row. The coordinate arrays describing the element's endpoints were separated into individual columns representing the start and end coordinates in three-dimensional space. Other than transforming the JSON structures into a tabular format that can be easily used by learning algorithms and statistical tools, this also makes the geometric properties of each structural element easier to analyze and interpret.

6. Cross-Section Encoding

```
for idx, bit in enumerate(cs_vector):  
    element_row[f"cross_section_cat_{idx}"] = bit
```

The cross-sectional category of each structural element was also expanded/flattened into multiple binary columns using one-hot encoding. This ensures that the cross-section category information is represented numerically in a format that machines can understand, without imposing a false ordinal relationship between categories.

7. Data Cleaning (Handling Missing Values)

```
element_row = {  
    ...  
    "if_beam": if_beam[i] if i < len(if_beam) else 0.0,  
    "if_roof_beam": if_roof[i] if i < len(if_roof) else 0.0,  
    "metal_deck_area": deck_area[i] if i < len(deck_area) else 0.0  
}
```

Before accessing each attribute array, a check was performed to verify if the requested index exists. If a value was missing, a default value of 0.0 was



assigned for it. This preserves the affected records while ensuring that the converted table contains a complete and consistent dataset. Not to mention that missing structural attributes can also correspond to the absence of that specific feature (e.g., no deck area means deck area = 0). Thus, retaining these records is preferable than discarding them entirely because their removal could reduce the amount of data available for analysis.

8. Graph Connectivity Extraction

```
for category, send_key, rcv_key in edge_types:
    senders = edges.get(send_key, [])
    receivers = edges.get(rcv_key, [])

    for idx in range(min(len(senders), len(receivers))):
        s_local = senders[idx]
        r_local = receivers[idx]

        all_edges.append({
            "connection_type": category, ... })
```

Structural connectivity information (how structural elements are linked together) was extracted from the graph representation of each building. From there, three connection types were recorded: (a.) column-to-column; (b.) beam-to-column; and (c.) column-to-ground. Each connection was stored as an independent connectivity record containing the sender element, receiver element, and connection type. These records were stored separately from the structural elements table to preserve the relational structure of the data and at the same time, support future graph-based analyses.

9. Story-Level Drift Ratio Extraction



```
for reverse_idx, drift_values_list in enumerate(reversed(drift_ratios)):
    story_row = {
        "source_file": file_name,
        "custom_building_id": custom_building_id,
        "story_level": reverse_idx, # Last entry in the JSON array becomes Story 0
    }

    for val_idx, val in enumerate(drift_values_list):
        story_row[f"drift_ratio_case_{val_idx}"] = val
```

The drift ratio values were extracted and reorganized into a separate story-level table where each row represents one story of a building, and each column represents a different loading case (flattened). The order of the original drift ratio list was reversed so that the zero-drift baseline corresponds to the ground floor (Story 0). This is because in structural simulations, the ground floor is typically anchored to the foundation, reason why it experiences little-to-no inter-story drift. Since the drift ratio is one of the important variables in structural analysis, it must be accurately associated with the correct building, story level, and element features.

10. Creating the Tables

```
df_elements = pd.DataFrame(all_elements)
df_edges = pd.DataFrame(all_edges)
df_stories = pd.DataFrame(all_story_analytics)

df_elements.to_csv("all_structural_elements.csv", index=False)
df_edges.to_csv("all_structural_edges.csv", index=False)
df_stories.to_csv("all_story_analytics.csv", index=False)
```

After that, all collected records were converted into pandas DataFrames and are then exported as CSV files. Three separate tables were produced in this conversion: (a.) Elements Table that contains geometric and



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physical attributes of each structural member; (b.) Connectivity Table that maps how structural elements are linked together; and (c.) Story-Level Analytics Table that contains story-level drift ratio measurements.

B. Data Reduction

This procedure was carried out by extracting only the most important attributes from the structural dataset, ensuring that the final representation remained concise yet meaningful. After converting the raw JSON files into tabular form, the dataset was filtered to retain buildings with complete structural grids, uniform spacing, and consistent story heights. From these validated structures, only the essential parameters were derived, specifically the building id, local element id, story level, end point location (start x, y, z), (end x,y,x) and drift ratio ex (x,y) drift ratio ey (x,y) for analyzing max IDR. These attributes were chosen because they capture the fundamental vertical and horizontal organization of a building, making them sufficient to describe its structural configuration. By focusing on these key features, the dataset was reduced into a compact form that preserves the critical geometric and spatial properties needed for analysis, while eliminating redundant or less relevant information.

C. Structural Parameter Extraction

This is a procedure performed on the structural dataset generated from the JSON-to-table conversion of the LSDSE dataset. The converted dataset contains 2,800 graph-based structural records, each representing a building and its associated structural elements. To ensure consistency and reliability in the extracted parameters, an instance selection process was applied to identify structures with complete rectangular grids, uniform horizontal spacing, and uniform vertical spacing. After applying these criteria, 423 buildings were retained for further



analysis. The selected instances were then subjected to feature extraction to derive a standardized set of structural parameters from their geometric and spatial properties. The resulting dataset serves as a reference for identifying realistic structural configurations and establishing plausible input ranges for the user-defined parametric inputs of the proposed system.

1. Instance Selection

Instance Selection identifies the subset of buildings that satisfy the geometric requirements necessary for parameter extraction. Since the objective is to derive a consistent parametric representation, structures exhibiting incomplete grids or irregular spacing are excluded. This process reduces the original dataset to a collection of buildings that share comparable structural characteristics and can therefore be represented using a common set of parameters.

Data Loading and Structural Element Selection

```
import pandas as pd
import numpy as np

FILE = "lsdse_json2table.xlsx"
SHEET = "structural_elements"
TOLERANCE = 0.01

df = pd.read_excel(FILE, sheet_name=SHEET, skiprows=[1])
columns = df[df["if_beam"] == 0].copy()
```

The structural dataset is loaded from the converted LSDSE spreadsheet and filtered to retain only column elements. Columns serve as the primary reference points for identifying structural grids and deriving geometric relationships within each building.



Building-Level Segmentation

```
results = []  
  
for building_id in sorted(columns["continuous_building_id"].unique()):  
    bldg = columns[columns["continuous_building_id"] == building_id]
```

Each building is processed independently by grouping records according to their building identifier. This prevents structural information from different buildings from being mixed during analysis.

Minimum Grid Requirement

```
x_lines = sorted(np.round(bldg["start_x"].astype(float).unique(), 3))  
y_lines = sorted(np.round(bldg["start_y"].astype(float).unique(), 3))  
  
if len(x_lines) < 2:  
    continue  
  
if len(y_lines) < 2:  
    continue
```

The first validation step verifies that sufficient horizontal coordinate information exists to define a structural grid. Buildings lacking adequate coordinate data cannot be represented using the intended parameter set and are therefore excluded.

Rectangular Grid Validation

```
coords = set(zip(  
    np.round(bldg["start_x"].astype(float), 3),  
    np.round(bldg["start_y"].astype(float), 3)  
))  
  
expected = len(x_lines) * len(y_lines)  
actual = len(coords)
```



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```
if actual != expected:  
    continue
```

The structural layout is evaluated to determine whether all expected grid intersections are present. Buildings that do not form complete rectangular grids are excluded from further processing.

Uniform Horizontal Spacing Validation

```
x_spans = np.diff(x_lines)  
y_spans = np.diff(y_lines)  
  
if np.std(x_spans) > TOLERANCE:  
    continue  
  
if np.std(y_spans) > TOLERANCE:  
    continue
```

The spacing between adjacent grid lines is examined to determine whether the structural layout maintains consistent horizontal intervals. Buildings with irregular spacing are excluded.

Vertical Level Identification and Validation

```
z_values = sorted(set(  
    np.round(  
        pd.concat([bldg["start_z"], bldg["end_z"]]).astype(float),  
        3  
    )  
))  
  
if len(z_values) < 2:  
    continue  
  
story_heights = np.diff(z_values)
```



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```
if np.std(story_heights) > TOLERANCE:  
    continue
```

The vertical configuration of each building is analyzed using elevation coordinates. Multiple elevation levels must exist, and floor-to-floor spacing must remain consistent throughout the structure.

2. Feature Extraction

Feature extraction derives structural parameters from the validated building instances by transforming geometric and spatial information into a standardized parametric representation. The resulting features provide a concise description of each building's structural configuration.

Horizontal Structural Parameter Extraction

```
bay_count_x = len(x_lines) - 1  
bay_count_y = len(y_lines) - 1  
  
bay_width_x = round(float(np.mean(x_spans)), 3)  
bay_width_y = round(float(np.mean(y_spans)), 3)
```

The validated horizontal grid information is used to derive structural parameters describing the horizontal arrangement of the building.

Vertical Structural Parameter Extraction

```
floor_count = len(z_values) - 1  
story_height = round(float(np.mean(story_heights)), 3)
```

The validated elevation information is used to derive parameters describing the building's vertical configuration.



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Parametric Record Construction

```
results.append({  
    "Building_ID": int(building_id),  
    "Floor_Count": floor_count,  
    "Story_Height": story_height,  
    "Bay_Count_X": bay_count_x,  
    "Bay_Width_X": bay_width_x,  
    "Bay_Count_Y": bay_count_y,  
    "Bay_Width_Y": bay_width_y  
})
```

The extracted parameters are consolidated into a single record representing the structural characteristics of one building.

3. Output

The resulting dataset contains the structural parameters extracted from all selected building instances. This parametric representation serves as a reference for analyzing structural configurations and establishing plausible ranges for user-defined inputs within the system.

Dataset Generation

```
output = pd.DataFrame(results)
```

The extracted records are assembled into a tabular dataset in which each row corresponds to a building and each column corresponds to a derived structural parameter.

Dataset Exportation

```
output.to_excel("regular_buildings.xlsx", index=False)
```



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The validated elevation information is used to derive parameters describing the building's vertical configuration. The resulting dataset contains one record for each building that satisfied all instance selection criteria. Each record consists of a standardized set of structural parameters describing the horizontal and vertical configuration of the building.

Parameters

Building_ID	Floor_Count	Story_Height	Bay_Count_X	Bay_Width_X	Bay_Count_Y	Bay_Width_Y
-------------	-------------	--------------	-------------	-------------	-------------	-------------

The extracted dataset serves as a condensed representation of the original structural data while preserving the geometric characteristics relevant to structural configuration. Floor Count and Story Height describe the building's vertical organization, whereas Bay Count X, Bay Count Y, Bay Width X, and Bay Width Y describe its horizontal grid arrangement. Collectively, these parameters provide a structured reference for analyzing structural patterns within the dataset and establishing plausible parameter ranges for user-defined parametric inputs in the proposed system.

III. Metric and Its Extraction

The primary performance metric used in this study is the Maximum Interstory Drift Ratio (MaxIDR), defined as the largest relative lateral displacement between two adjacent floors divided by the story height, taken as the absolute maximum across all story levels and all seismic load directions for a given structure.

The structural performance metrics identified for this study include Peak Stress, Peak Deformation, Peak Displacement, and Maximum Inter-story Drift Ratio (MaxIDR). However, Peak Stress, Peak Deformation, and Peak Displacement cannot be utilized in the



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present analysis because these parameters are not available within the current dataset and can only be obtained directly from Non-Linear Time History Analysis (NLTHA) seismic simulation outputs. Since the dataset contains only post-processed drift response information, these metrics cannot be derived or reconstructed from the available attributes. In contrast, MaxIDR can be directly extracted from the four recorded drift ratio components (drift_ratio_ex_x, drift_ratio_ex_y, drift_ratio_ey_x, and drift_ratio_ey_y), which represent structural responses under bidirectional seismic excitation. Consequently, MaxIDR serves as the only performance metric that can be consistently derived from the dataset, making it the most appropriate response variable for the present analysis.

MaxIDR is also well-established in seismic codes as a damage-sensitive indicator. It correlates directly with structural and non-structural damage levels and is the basis for drift limits in the National Structural Code of the Philippines (NSCP) and ASCE 7.

1. Data Loading and Drift Ratio Selection

```
import pandas as pd
from openpyxl import load_workbook

INPUT = r'C:\Users\Zyvene\Downloads\lsdse-idr.xlsx'
OUTPUT = r'C:\Users\Zyvene\Downloads\lsdse-idr-summary.xlsx'
THRESHOLD = 0.015

drift_cols = ['drift_ratio_ex_x', 'drift_ratio_ex_y',
              'drift_ratio_ey_x', 'drift_ratio_ey_y']

df = pd.read_excel(INPUT)
```

The IDR dataset is loaded from the LSDSE spreadsheet and the four drift ratio components are selected for analysis. These attributes represent story-level responses under bidirectional seismic excitation and serve as the basis for computing the Maximum Inter-story Drift Ratio (MaxIDR).



2. Drift Ratio Normalization and Filtering

```
df[drift_cols] = df[drift_cols].abs()  
df = df[df[drift_cols].max(axis=1) > 0]  
  
df['row_max'] = df[drift_cols].max(axis=1)
```

The drift ratio values are converted to their absolute magnitudes to ensure that drift responses are evaluated independent of direction. Records containing zero drift values are excluded, and the maximum drift ratio for each story is computed from the four seismic response components.

3. Building-Level MaxIDR Extraction

```
summary = (  
    df.groupby('custom_building_id')  
    .agg(  
        Max_IDR = ('row_max', 'max'),  
        Num_Floors = ('story_level', 'max'),  
    )  
    .reset_index()  
)
```

The dataset is grouped according to building identifiers to derive structure-level performance metrics. For each building, the maximum drift ratio observed across all stories is extracted as the MaxIDR, while the highest story level is used to determine the total number of floors.

4. Critical Story Identification



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```
crit = (  
    df.loc[df.groupby('custom_building_id')['row_max'].idxmax(),  
           ['custom_building_id', 'story_level']]  
    .rename(columns={'story_level': 'Critical_Story'})  
)  
  
summary = summary.merge(crit, on='custom_building_id')
```

The story level corresponding to the maximum drift ratio is identified for each structure. This value is recorded as the Critical Story, representing the floor where the greatest relative lateral displacement occurs.

5. Performance Classification

```
summary['Status'] = summary['Max_IDR'].apply(  
    lambda v: 'FAIL' if v > THRESHOLD else 'PASS'  
)
```

Each structure is evaluated against the allowable drift threshold of 0.015. Buildings exceeding this limit are classified as FAIL, while those within the allowable range are classified as PASS.

6. Dataset Construction

```
summary['Structure'] = summary['custom_building_id'] + 1  
  
summary = summary[['Structure',  
                   'Num_Floors',  
                   'Critical_Story',  
                   'Max_IDR',
```



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'Status']]

The extracted performance metrics are consolidated into a standardized summary table. Each record represents a single building and contains the number of floors, critical story, maximum drift ratio, and corresponding performance classification.

7. Output Generation

```
wb = load_workbook(INPUT)

if 'IDR Summary' in wb.sheetnames:
    del wb['IDR Summary']

ws = wb.create_sheet('IDR Summary', 0)

headers = ['Structure',
           'No. of Floors',
           'Critical Story',
           f'Max IDR (Threshold={THRESHOLD})',
           'Status']

ws.append(headers)

for row in summary.itertuples(index=False):
    ws.append([
        row.Structure,
        row.Num_Floors,
        row.Critical_Story,
        row.Max_IDR,
        row.Status
    ])

wb.save(OUTPUT)
```



The extracted metrics are written to a new worksheet named IDR Summary. The resulting dataset provides a building-level representation of seismic drift performance and serves as the primary source for the subsequent statistical and exploratory analyses.

IV. Initial Findings and Data Interpretation

Performance Classification

A threshold of 0.015 (1.5%) is applied, consistent with allowable story drift limits for moderate-ductility building systems:

$$\text{Status} = \text{PASS if } \text{MaxIDR} \leq 0.015 \mid \text{FAIL if } \text{MaxIDR} > 0.015$$

Results

Out of 2,800 structures evaluated from the dataset:

Classification	Count	Percentage
PASS ($\text{MaxIDR} \leq 0.015$)	20	0.71%
FAIL ($\text{MaxIDR} > 0.015$)	2,780	99.29%
Total	2,800	100%

The overwhelming majority of structures exceed the allowable drift threshold, indicating widespread seismic vulnerability across the evaluated building inventory. The high failure rate indicates that most structures in the dataset exceeded the selected drift threshold under the analyzed loading conditions.



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Relationship Between Structure Variables

To understand the interaction between the number of floors, the critical story, and the maximum IDR, we can look at the following trends:

- **Max IDR:** The maximum IDR is the most important performance indicator because it directly determines whether a structure satisfies the allowable drift requirement. Any structure with a maximum IDR exceeding the threshold of 0.015 is classified as a failure regardless of the performance of the remaining stories. Based on the dataset, the average maximum IDR is 0.02387, which exceeds the allowable limit by approximately 59.15%, indicating that excessive drift is a common characteristic among the analyzed structures.

The results further show that 2,780 out of 2,800 structures (99.29%) failed the drift criterion, while only 20 structures (0.71%) achieved a passing classification. Additionally, all floor categories recorded average maximum IDR values above the allowable threshold, demonstrating that excessive drift is present across the entire dataset rather than being limited to specific building heights. These findings emphasize the importance of identifying the critical story and implementing measures to reduce lateral displacement, as controlling the maximum IDR is essential for improving the overall seismic performance of the structure.

- **Critical Story:** The critical story refers to the level where the maximum inter-story drift ratio occurs. The data reveals a very strong positive relationship between the number of floors and the critical story, with a correlation coefficient of $r = 0.986$ and a coefficient of determination ($R^2 = 0.972$). This indicates that the location of



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maximum drift follows a highly consistent pattern and is strongly associated with building height.

The results show that 2,454 out of 2,800 structures (87.64%) recorded their critical story at the highest floor, while 2,723 structures (97.25%) exhibited the critical story either at the top floor or one floor below it. This concentration of critical stories near the upper levels suggests that lateral displacement demands accumulate toward the top of the structure, making these stories more susceptible to excessive drift. Consequently, improving seismic performance should focus on enhancing stiffness and displacement control near the upper portions of the building, where peak drift demands are most frequently observed.

- **Number of Floors and IDR:** The relationship between the number of floors and the maximum Inter-story Drift Ratio (IDR) was examined to determine whether increasing building height significantly influences seismic drift performance. Based on the dataset, the number of floors does not appear to be a dominant factor affecting maximum IDR values. Although a positive correlation coefficient of $r = 0.362$ was observed between the number of floors and the maximum IDR, this relationship is considered relatively weak to moderate. The corresponding coefficient of determination ($R^2 = 0.131$) indicates that only approximately 13.1% of the variation in maximum IDR can be explained by differences in building height alone. Consequently, nearly 86.9% of the observed variation is attributed to other factors not represented by the floor count variable, such as structural stiffness distribution, member properties, geometric configurations, or load-resisting system characteristics.



To further examine this relationship, the structures were grouped according to the number of floors, and the average maximum IDR was computed for each category, as shown in the table below.

Number of Floors	Number of Structures	Average Maximum IDR
1	264	0.02021
2	294	0.02238
3	267	0.02285
4	274	0.02406
5	287	0.02437
6	319	0.02495
7	289	0.02483
8	258	0.02450
9	256	0.02518
10	292	0.02514
Total	2,800	0.02387

The results show a general increase in average maximum IDR as the number of floors increases; however, the increase is relatively small compared to the increase in building height. The average maximum IDR rises from 0.02021 for one-story structures to 0.02514 for ten-story structures, representing only a 24.4% increase despite a 900% increase in the number of floors. Furthermore, the trend is not strictly linear, with slight decreases observed between six-, seven-, and eight-story structures, indicating that drift performance does not consistently worsen as height increases.



All floor categories recorded average IDR values above the allowable limit of 0.015, with one-story structures exceeding the threshold by approximately 34.7% and ten-story structures exceeding it by approximately 67.6%. However, the correlation coefficient ($r = 0.362$) and coefficient of determination ($R^2 = 0.131$) indicate that the number of floors explains only about 13.1% of the variation in maximum IDR. This suggests that while building height contributes to drift behavior, other structural characteristics have a much greater influence on seismic performance than floor count alone.

V. Ethics

The dataset used in this study is the LSDSE Dataset developed by Kai-Hung Chang and Chin-Yi Cheng and publicly released for research purposes as part of the study Learning to Simulate and Design for Structural Engineering (LSDSE). The dataset consists of structural engineering data and simulation outputs that do not contain personally identifiable information, sensitive personal records, or confidential information related to individuals. As a result, the use of the dataset does not present concerns regarding privacy, informed consent, or personal data protection that are commonly encountered in human-centered datasets.

The dataset was obtained from its official public source and used solely for academic and educational purposes consistent with its original research intent. Proper attribution is provided to the original authors and source of the dataset to acknowledge their contribution and to maintain transparency in the use of publicly available research data. No attempt was made to claim ownership of the dataset or its original findings.



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Throughout the preprocessing stage, data reduction, instance selection, and feature extraction were performed to prepare the dataset for analysis. These procedures were conducted in a transparent and systematic manner without altering, fabricating, or misrepresenting the original structural information. Buildings that did not satisfy the established validation criteria were excluded based on clearly defined geometric requirements rather than subjective selection, helping ensure consistency and reproducibility of the results.

The findings of this study are interpreted within the scope and limitations of the selected dataset. Any observed patterns or trends are presented as characteristics of the analyzed structural records and should not be generalized beyond the dataset without further validation. By maintaining accurate data handling practices, documenting preprocessing decisions, and properly acknowledging the original source, the study seeks to uphold the principles of research integrity, transparency, and responsible data use.

VI. Conclusion

The data preprocessing and feature extraction procedures transformed the original LSDSE structural dataset into a standardized and analyzable format. Through data cleaning, instance selection, and parameter extraction, the original collection of 2,800 graph-based structural records was processed to identify 423 buildings with complete rectangular grids and consistent geometric properties. From these validated structures, key parametric attributes, including floor count, story height, bay count, and bay width, were extracted to create a condensed dataset suitable for subsequent analysis.

Using the Maximum Inter-story Drift Ratio (MaxIDR) as the primary performance metric, the analysis revealed that the overwhelming majority of the evaluated structures exceeded the allowable drift limit of 0.015, resulting in a failure rate of approximately



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99.29%. The results further showed that while the number of floors exhibits only a weak-to-moderate relationship with MaxIDR, the critical story consistently occurs near the uppermost levels of the structure. This suggests that excessive lateral displacement is a common characteristic across the evaluated buildings and that peak drift demands tend to concentrate near higher elevations. Overall, the findings indicate that structural stiffness and displacement control remain critical considerations for improving seismic performance, while also demonstrating the effectiveness of data preprocessing and feature extraction techniques in uncovering meaningful structural patterns within large-scale engineering datasets.

VII. References

Chang, K.-H., & Cheng, C.-Y. (2020). LSDSE dataset [Data set]. Autodesk Research. <https://github.com/AutodeskAILab/LSDSE-Dataset>

Chang, K.-H., & Cheng, C.-Y. (2020). Learning to simulate and design for structural engineering. In H. Daumé III & A. Singh (Eds.), Proceedings of the 37th International Conference on Machine Learning (Vol. 119, pp. 1426–1436). PMLR. <https://proceedings.mlr.press/v119/chang20a.html>

Gantt Chart:

[illegible]

	Task Title	Task Owner	Start Date	Due Date	Days
1	Acquiring/ Finding Data Set	ALL	2/23/26	5/2/26	69
	Initial Presentation the Dataset	ALL	3/7/26	3/7/26	
2	Determining Data Analysis Technique Suitable for Research	ALL	5/2/26	5/5/26	3
3	Data Mining Activity		5/5/26	5/30/26	25
	a. JSON to Excel Table	Micaela Madayag	5/5/26	5/30/26	25
	b. Data Transformation	Micaela Madayag	5/5/26	5/30/26	25
	c. Data Cleaning	Micaela Madayag	5/5/26	5/30/26	25
	d. Metric Extraction	Zyvene	5/5/26	5/30/26	25
	e. Data Reduction	Hinnah Guijaro	5/5/26	5/30/26	25
	e. Thesis Parametric Inputs	Sean Padilla	5/5/26	5/30/26	25
4	Documentation	All	5/30/26	6/5/26	5
	I. The Dataset	Hinnah Guijaro	5/30/26	6/5/26	5
	II. Dataset Preparation		5/30/26	6/5/26	5
	a. Conversion of the Dataset from .json File to Table	Micaela Madayag	5/30/26	6/5/26	5
	b. Data Reduction	Hinnah Guijaro	5/30/26	6/5/26	5
	c. Structural Parameter Extraction	Sean Padilla	5/30/26	6/5/26	5
	III. Metric and It's Extraction	Zyvene Guadalupe	5/30/26	6/5/26	5
	IV. Initial Findings and Data Interpretation	Zyvene Guadalupe Hinnah Guijaro	5/30/26	6/5/26	5
	V. Conclusion	Sean Padilla	5/30/26	6/5/26	5
Final Presentation			6/8/26		



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Contribution List			
Name	Contribution	Percentage of Work	Score
Guadalupe, Zyvene D.	Data Mining Activity • Metric Extraction Documentation • Metric and Its Extraction • Initial Findings and Data Interpretation	25%	5/5
Guijaro, Hinnah Lyka L.	Data Mining Activity • Data Reduction Documentation • The Dataset • Dataset Preparation: Data Reduction • Initial Findings and Data Interpretation	25%	5/5
Madayag, Micaela C.	Data Mining Activity • Conversion of JSON Files to Excel Tables ◦ Data Transformation ◦ Data Cleaning Documentation • Dataset Preparation: Conversion of the Dataset from JSON Files to Tabular Format	25%	5/5
Padilla, Sean Timothy B.	Data Mining Activity • Thesis Parametric inputs Documentation	25%	5/5



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Contribution List			
	• Dataset Preparation: Structural Parameter Extraction • Conclusion		